

Benchmarking Sustainable Raw Material Alternatives Without a Sustainability Score

Eligibility, comparability, evidence and trade-off: a decision method demonstrated on a synthetic chemical-procurement dataset and an analytics prototype

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Abstract

Procurement organisations are increasingly asked to compare raw-material alternatives on cost and carbon at the same time. The difficulty is no longer calculating a product carbon footprint; it is turning one into something a sourcing decision can use. The comparison fails in three specific ways: options are compared that cannot actually be bought, numbers are compared that were never calculated on the same basis, and dimensions are merged that should have stayed apart. None of these failures is solved by a better score.

This paper develops a decision method organised as a sequence — establish *eligibility*, then establish *comparability*, then show the *trade-off*, with evidence assessed before the trade-off is put to a named decision owner. Ten success dimensions give the decision context a structured view, each made assessable through measurement indicators and read against a benchmark rather than combined into one figure. The method is supported by a four-role convention that separates gates from performance, evidence and context; by an operationalisation of the six comparability conditions of the GHG Protocol *Product Standard*, Appendix A §A.1; and by an evidence model that keeps provenance, data quality, verification and uncertainty outside the performance number.

The method is demonstrated on a synthetic dataset of 25 material alternatives across five independent benchmark cases, carrying deliberately embedded data-quality defects, harmonised through a deterministic pipeline into an audited semantic layer of 21 measures and displayed through a working analytics prototype. Four findings are named: the portfolio total is overstated approximately fivefold when mutually exclusive scenarios are summed; the two options with the strongest cost and carbon position in one case are both technically blocked; the strongest carbon result in another case rests on a 2019 industry-average factor; and one material has no carbon figure at all and is never assigned one.

The paper states plainly what the demonstrator does *not* establish. No §A.1 comparability condition is fully satisfied by it today, and the eight fields that would close the gap are named. All data is synthetic and labelled as such. No sustainability score, weighting or cross-case ranking is produced at any point.

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1 Introduction

Raw-material sourcing decisions have acquired a second objective. Where a category manager once had to reconcile price, quality and security of supply, the same decision is now also expected to deliver a carbon reduction, and to withstand questioning about water, circularity, hazardous substances and supply concentration. A common organisational response is to add columns to a supplier scorecard.

That response misdiagnoses the problem. Adding columns assumes the difficulty is a shortage of dimensions. It is not. The difficulty is that the dimensions behave differently in a decision, that the numbers placed in those columns are frequently not comparable with one another, and that a large share of the cells are empty in a way that a scorecard conceals rather than reports.

This paper takes the opposite route. Instead of enriching a score, it removes the score and replaces it with a sequence:

ELIGIBILITY → COMPARABILITY → EVIDENCE
→ TRADE-OFF → DECISION AND GOVERNANCE

An alternative must first be shown to be *eligible* — legally and technically capable of being bought. Its figures must then be shown to be *comparable* with the incumbent's, and the *evidence* behind those figures must be assessed rather than assumed. Only then is the remaining *trade-off* put in front of a named decision owner, who records the decision and what was given up. Each step has a failure mode that is invisible if the steps are collapsed.

The argument is developed on a synthetic demonstrator rather than on industrial data. This is a deliberate constraint and it has a cost, discussed in Section 14. It also has a benefit: because the dataset was authored, its data-quality defects are known exactly, and the method can be shown handling them rather than asserted to handle them.

Contributions. The main contributions are:

1. a decision sequence that makes each failure mode visible before the next one is judged;
2. ten success dimensions, each made assessable through measurement indicators and read against a benchmark rather than aggregated;
3. a four-role convention — gate, performance, evidence, context — presented explicitly as a project synthesis rather than a validated framework;
4. an operationalisation of the six §A.1 comparability conditions as a three-state verdict usable inside a sourcing decision;
5. an evidence model separating provenance, data quality, verification and uncertainty from the performance number, with missing values preserved as missing at four independent layers;

6. a reproducible synthetic demonstrator, and an explicit statement of what it does not establish together with the fields that would close the gap.

2 Background and related work

2.1 What the standards settle

The methodological base for product-level carbon accounting is mature. The GHG Protocol *Product Life Cycle Accounting and Reporting Standard* [1] defines how a product carbon footprint (PCF) is calculated, and ISO 14040/14044 with the associated pedigree guidance [7] define what data quality means in a life-cycle inventory. For the chemical sector, the *Together for Sustainability* PCF guideline [12] and the cross-sector PACT methodology [3, 13] specify how a supplier-declared footprint should be produced and exchanged; the Catena-X rulebook [14] does the same for automotive value chains. ISO 20400 [6] places sustainability criteria alongside price and quality inside a single procurement evaluation rather than in a parallel process.

Beyond climate, methods also exist and are frequently assumed not to. ISO 14046 [24] defines a product water footprint including consumption and degradation, with mandatory attention to local hydrological conditions. The EU Product Environmental Footprint, recommended for use by Commission Recommendation (EU) 2021/2279 [25], covers sixteen impact categories including toxicity, land use and resource depletion, operationalised per product group through category rules.

The regulatory frame adds duties rather than decision rules: ESRS topical standards E1–E5 [9], the Critical Raw Materials Act [8], the Carbon Border Adjustment Mechanism [11], the Ecodesign for Sustainable Products Regulation [10], and corporate sustainability due diligence as amended by Directive (EU) 2026/470 [5, 22]. REACH [23, 4] defines specific prohibitions and specific information duties, which — as Section 5.1 shows — are frequently confused with one another.

2.2 What the standards do not settle

Three gaps matter for a sourcing decision.

First, the carbon standard limits itself explicitly. §1.7 of [1] sets out that a climate-only inventory can miss trade-offs and co-benefits across other environmental impacts, and directs that its results **not be treated as a statement of a product's overall environmental performance**.¹

Second, the same standard states that it does not, by itself, support the comparison a buyer wants to make. Appendix A distinguishes the intended applications; performance tracking is supported as published, while consumer and business purchasing decisions require additional programme specifications or

¹The standard names ecosystem degradation, resource depletion, ozone depletion and negative human health impacts among the categories at risk [1].

product-specific guidance. Section 5.2 makes this operational.

Third, the literature on decision method is thin where it matters most. Material selection under constraints is well established [18, 17], and life-cycle data quality has an established vocabulary [7, 19], but neither tells a category manager what to do when the best-performing candidate is not approved, or when the best carbon figure has the weakest provenance. That gap is where this paper sits.

3 Success dimensions and measurement indicators

A sourcing decision is easier to argue when the dimensions it turns on are named before the numbers arrive. Success dimensions provide a structured view of the decision context. Suitable measurement indicators make each dimension assessable against a relevant benchmark. The reference point may be a target level, a peer profile or — as in the demonstrator from Section 4 onward — the incumbent. Figure 1 illustrates how such a multidimensional benchmark profile can be visualised without aggregating the dimensions into a single score.

success dimensions → measurement indicators
→ benchmark comparison → decision support

Table 1 lists ten dimensions that recur in raw-material sourcing, with examples of the indicators that make them assessable. Which dimensions are relevant, and which indicators are suitable, depends on the material, the application and the decision context.

Reading the profile. The value of the profile is in its shape. A profile that meets the benchmark on eight dimensions and falls two levels short on one is a different decision from a profile that is uniformly one level short, even though a single combined figure would treat them alike. Keeping the dimensions apart is what preserves that difference.

4 Problem formulation

4.1 Comparison cases and the incumbent baseline

A benchmark exercise is organised into *cases*. A case c is one purchased material need, containing exactly one incumbent $b(c)$ — the material currently bought — and a set of candidate alternatives. Cases are independent: an alternative in one case is not substitutable for an alternative in another, so no statement is made across cases.

Within a case, performance is expressed relative to the incumbent. For an alternative i in case c , with annual spend s_i and annual emissions g_i ,

$$\Delta_i^{\text{cost}} = \frac{s_i - s_{b(c)}}{s_{b(c)}}, \quad \Delta_i^{CO_2e} = \frac{g_i - g_{b(c)}}{g_{b(c)}}. \quad (1)$$

Two properties of Equation (1) carry the method. First, both deltas are computed from absolute quantities, not from stored percentages, so a rounded column can never enter a decision. Second, $\Delta_i^{CO_2e}$ is *undefined* where g_i or $g_{b(c)}$ is unavailable. It is not zero, and it is not imputed. An undefined carbon delta removes that material from the carbon comparison and from nothing else.

4.2 Three failure modes

Figure 2 states the sequence. Each stage exists because a specific failure is otherwise invisible.

Eligibility asks whether an alternative can be considered at all. The failure is that an option is compared, and often preferred, that cannot legally or technically be bought. The control is to screen on true constraints first and to keep what was screened out visible, so that the screen can be argued with.

Comparability asks whether two figures were ever comparable. The failure is that two numbers produced under different rules are subtracted from one another. The control is to establish equivalence of function, boundary, allocation, period and assurance *before* the subtraction, not after.

Trade-off asks what is being given up and who owns that decision. The failure is that one dimension improves while another quietly moves the wrong way. The control is to hold the dimensions on separate axes and to name the person who accepts the residual.

5 The decision method

The method has five steps — eligibility, comparability, evidence, trade-off, and a recorded decision with an owner. The first four are treated in this section; the fifth is Section 11. The steps are ordered, and the ordering is the point: a later step cannot repair an earlier one.

5.1 Four decision roles

Dimensions that look alike on a data sheet behave completely differently in a decision. Four behaviours are distinguished, summarised in Table 2. **This four-role convention is a project synthesis**, not a validated external framework: constraint-versus-objective thinking is supported by materials-selection method [18], evidence metadata by [7, 3], and legal duties by law, but no source defines the complete four-part architecture.

The essential refinement is that *a factor’s role is set by the decision rule, not by the topic name*. Supply risk is a gate where a minimum assured volume or a dual-source policy is contractual, a performance dimension where lower concentration is merely preferred, and context where country of origin only explains exposure. Regulatory criteria are gates only where the specific use is unlawful, unauthorised, or outside a written company policy; disclosure duties are not gates. Social criteria are gates for prohibited conduct or an explicit minimum requirement, while general due diligence is a risk-based process [5, 22, 15],

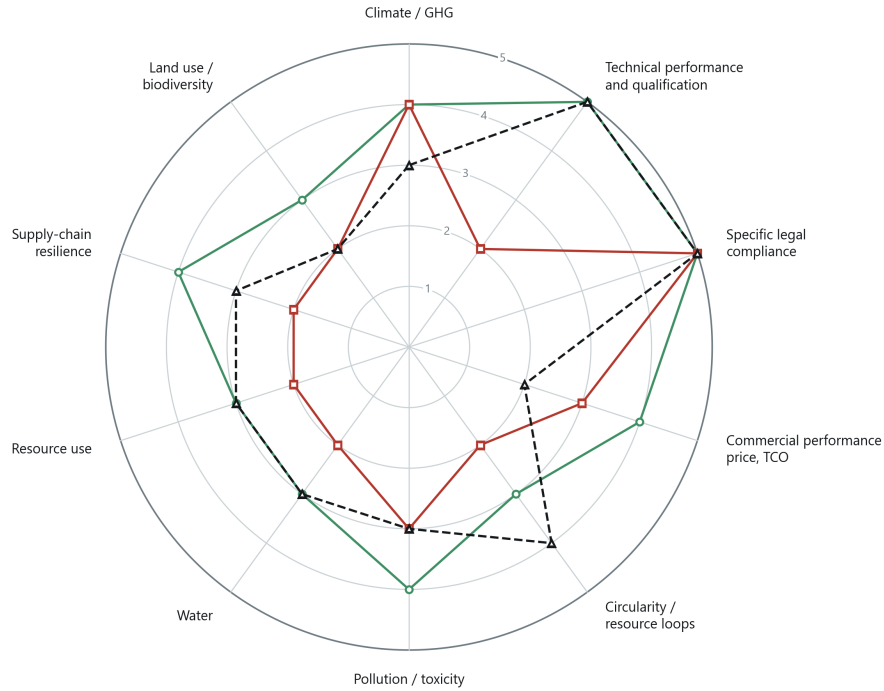


Figure 1. A multidimensional benchmark profile. Ten success dimensions on a common five-level scale, each read against a benchmark rather than combined with the others. Green marks the benchmark or target level, red the observed profile, black a second profile for comparison. **The values shown are illustrative examples, not empirical assessments.** The dimensions stay separate: no weighting is applied, no total is formed and no ranking is derived.

not a certificate.

A practical consequence: every gate must name a threshold owner, a decision owner and a time horizon. A gate without a named owner is an opinion.

What is not a gate. Presence of a substance on the REACH Candidate List is a disclosure duty, not a prohibition on placing a product on the market [23]. General corporate due-diligence status is a risk-based process obligation applying from 2029 to a narrow set of very large undertakings [5, 22], not a supplier pass mark. Treating either as a gate removes candidates that are lawful to buy; treating an actual prohibition as a scoring input keeps candidates that are not.

5.2 Comparability: six conditions

Two supplier PCFs are not comparable because both are expressed in kg CO₂e per kg. Appendix A §A.1 of [1] establishes what a business purchasing comparison additionally requires. Table 3 operationalises those requirements as six checks for this method. A PCF value on its own satisfies none of them.

This reframes the sector rulebooks. TfS [12], PACT [3, 13] and Catena-X [14] are precisely the “additional programme specifications” the appendix requires. Naming one in a request for quotation is not bureaucracy; it is the condition that makes the subsequent comparison legitimate. Conversely, condition 3 breaks easily between two entirely honest suppliers: a producer that discloses use of economic allocation

where physical data is unavailable [20] is not doing anything wrong, but its figure is not subtractable from one built on physical allocation.

Because conformance is rarely complete, the underlying research derives a three-state verdict rather than a binary one: *comparable*, *conditionally comparable*, *not comparable*. The operative rule is that where the methodological gap could plausibly exceed the measured difference, **no difference is reported**. This is a source-supported interpretation, not a provision of the standard.

5.3 Evidence discipline

Evidence quality is routinely treated as one attribute. The structure used operationally throughout this paper separates four, each supported by the cited sources (Figure 3): *provenance* — where the number came from; *data quality* — how representative it is in time, geography and technology [7]; *verification* — whether anyone independent checked it; and *uncertainty* — how far apart two figures could actually be [1]. None of the four may be blended into the performance number. A quality-adjusted PCF destroys both the measurement and the evidence statement.

Two distinctions are easy to lose. *Supplier-specific* and *primary* are not synonyms: a supplier-specific footprint normally combines primary activity data with secondary background factors — which is exactly why PACT reports a *share* of primary data rather than a flag [3, 13]. And an industry-average emis-

Table 1. Ten success dimensions that recur in raw-material sourcing, with examples of the measurement indicators that make each one assessable. The class column records how far a dimension enters this project’s scope, not its decision role: A core, B context-dependent, C acts as a gate condition, D relevant to the wider picture but outside this prototype. Which dimensions are relevant, and which indicators are suitable, depends on the material, the application and the decision context.

	Success dimension	Class	Example measurement indicators
1	Climate / GHG	A	PCF in kg CO ₂ e per kg; CO ₂ e delta vs incumbent; PCF coverage; evidence attributes
2	Technical performance and qualification	C	Approval status; equivalence against the technical specification; qualification stage
3	Specific legal compliance	C	Authorisation or restriction status for the specific use; company policy thresholds
4	Commercial performance — price, TCO	A	Price per kg; annual spend; cost delta vs incumbent; qualification and switching cost
5	Circularity / resource loops	B	Recycled and renewable content; chain-of-custody model; retention of value
6	Pollution / toxicity	B	PEF toxicity categories; substance profile for the actual use and exposure
7	Water	B	ISO 14046 water footprint; basin-level water stress at the production site
8	Resource use	B	PEF resource-use categories; feedstock intensity
9	Supply-chain resilience	B	Awarded-volume concentration; dual-source status; lead time and recovery time
10	Land use / biodiversity	D	Land-occupation and transformation by origin; feedstock-specific pressure

Table 2. The four decision roles. Confusing them is a common structural error in supplier benchmarking, because it allows a strong performance number to compensate for a failed constraint, or an evidence weakness to be silently priced into a result. *Project synthesis*; the legal statements within it are fact.

Role	The question it answers	How it behaves	Examples
Gate	Can this alternative be considered at all?	Pass or fail. Never traded off	Technical approval; an Annex XIV sunset date without authorisation; an Annex XVII restriction covering the specific use
Performance	How does an eligible alternative perform?	Compared, plotted, traded off	PCF; price; annual spend
Evidence	How much confidence does that comparison deserve?	Reported beside the number, never blended into it	Provenance; data quality; verification; uncertainty
Context	Under what material, application and supply conditions?	Filters and explains. Not optimised	Material group; physical form; country of origin

sion factor, however recent, cannot by construction distinguish two suppliers operating within the same industry.

Missing is never zero. A missing environmental figure is a reportable state, not a value. Substituting zero inverts the incentive structure of the entire exercise: the supplier that discloses least obtains the best apparent result. In the demonstrator this rule is enforced at four independent layers — in the measure definitions, in conditional formatting, in the figure-generation code and in the absence of any imputation step in the pipeline — so that no single change can quietly reintroduce zero-filling.

5.4 Materiality beyond carbon

Not every dimension matters for every material. A universal environmental data request produces effort rather than decisions. The organising question is ma-

teriality: *could this dimension materially change the decision for this material, from this feedstock, from this origin?*

The answer varies by driver, not by fashion. Water impacts are basin-specific, so a generic per-kg water league table is methodologically wrong even where data exists [24]. Land use and biodiversity pressure attach to feedstock and origin rather than to the delivered kilogram. Circularity outcomes depend on material, feedstock and application [2]. Climate is the exception that proves the rule: it is in scope for every material in this project’s scope, which is why it is the dimension the prototype measures. Everything else enters the decision only where the screening question above returns yes for the specific material, feedstock and origin; the class column of Table 1 records where each dimension sits for the material groups in this project.

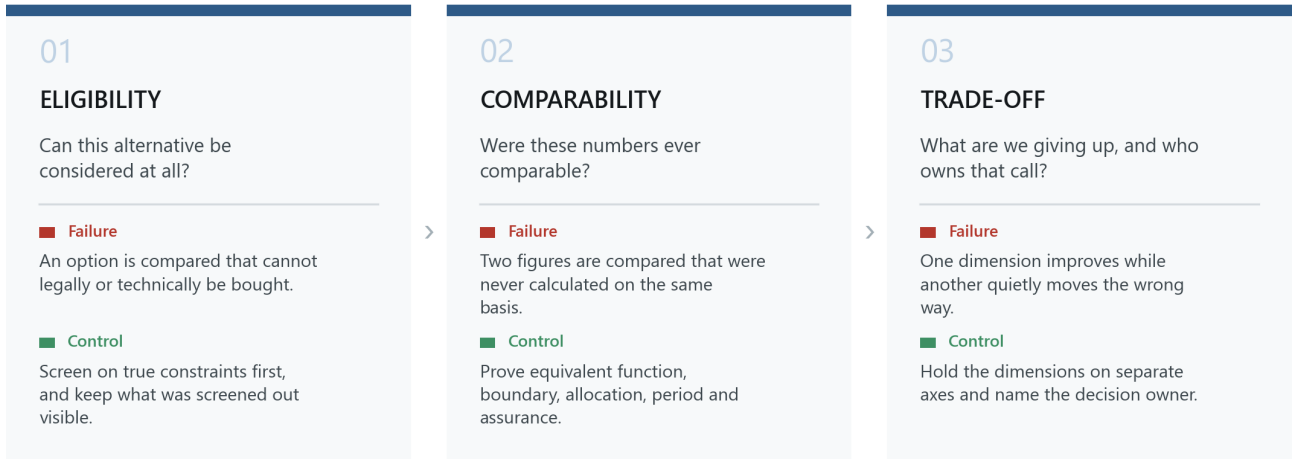


Figure 2. The three failure modes and the sequence that answers them. Each stage states the question it settles, the failure that occurs when it is skipped, and the control that prevents it. The sequence is not a workflow diagram: it is an ordering constraint, and its value is that a later stage may not repair an earlier one.

Table 3. Six comparison checks used by this method, with the practical failure that can defeat each. Operationalised by the author based on the comparability requirements of the GHG Protocol *Product Life Cycle Accounting and Reporting Standard*, Appendix A §A.1 [1]; the wording of the checks is the author's and implies no endorsement by the GHG Protocol, WRI or WBCSD. Conformance is declared per comparison, and where a check is unmet the comparison is reported as conditional rather than presented as a result.

	Comparison check	What defeats it in practice
1	Same unit and functional basis	A per-kg comparison of materials with different active content is a declared-unit comparison, not a functional one
2	Equivalent system and time boundaries	“Cradle-to-gate” alone does not establish which processes are included
3	Consistent allocation approach	Broken wherever one supplier allocates economically and another physically [20]
4	Comparable data-quality basis	Uncertainty is rarely reported, so close options cannot be distinguished
5	Comparable time and geographic representativeness	A reference year is not a data-collection period, and says nothing about geography
6	Appropriate independent assurance	The least often met in the sources reviewed

6 Practices and how well each is evidenced

Thirteen practices emerged from the underlying research, organised along the decision sequence and shown in Table 4. Four of them are strongly evidenced, resting on a primary standard or an EU authority read in the original. The remainder are supported by narrower citations, and one is an explicit project recommendation.

Showing which is which is itself a practice. A benchmark method that presents a plausible recommendation with the same authority as a standard requirement invites the reader to over-trust both. The three-state marker used in Table 4 describes only *how well a practice is supported*. It is not a maturity level, it does not describe an organisation, and no score is computed from it.

What a procurement organisation does differently, stated as compactly as the evidence allows: name the rulebook in the request for quotation; require the

primary-data share; keep evidence beside the number; never zero-fill a gap; and keep one decision record per case rather than two.

7 Synthetic dataset and harmonisation

7.1 Design

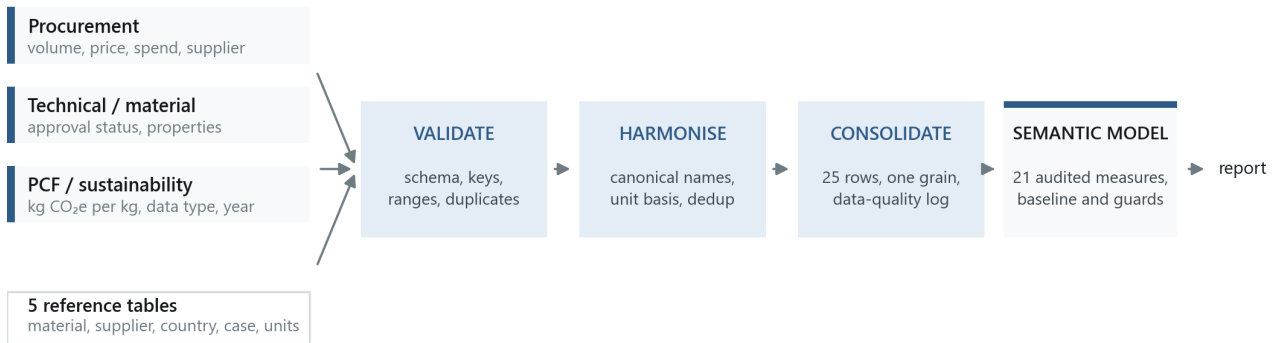
The dataset describes five benchmark cases — base oils, solvents, surfactants, polymer additives, resins and binders — each containing one incumbent and four alternatives, giving 25 material rows. It is authored, not sampled. Every value is synthetic, and no supplier, company or market data is used or implied anywhere in the project.

Authoring the data is what makes the demonstration honest rather than merely plausible. Because the defects were placed deliberately, the paper can show the method meeting them, and a reader can verify from the repository that nothing was cleaned away

BARE	QUALIFIED	ABSENT
2.4 kg CO ₂ e / kg	2.4 kg CO ₂ e / kg	— no figure reported
Provenance ?	Provenance supplier-specific; primary-data share	 never
Data quality ?	Data quality temporal, geographical, technological	
Verification ?	Verification third-party assurance performed	If absence defaults to zero, the least transparent supplier becomes the best performer.
Uncertainty ?	Uncertainty assessed and reported with the value	
A number. Not a measurement.	The same number, now usable.	Missing stays missing.

Figure 3. The same value in three states. *Bare*: a figure without the relevant evidence attributes is not yet usable in a comparison. *Qualified*: the same figure with provenance, data quality, verification and uncertainty attached becomes usable in a comparison. *Absent*: no figure was reported, and the cell stays empty — because if absence defaults to zero, the least transparent supplier becomes the best performer. The value shown is illustrative and is not taken from the dataset.

SOURCE SYSTEMS



Ground truth is held out of the transform and used only by an independent validation step. Eight data-quality findings are logged; two are left open by design.

Figure 4. Data lineage of the demonstrator. Three heterogeneous source extracts and five reference tables are validated, harmonised and consolidated into a single 25-row analytical grain, then exposed through an audited semantic layer. Ground truth is deliberately held out of the transform and read only by an independent validation step, so that the transform cannot be tuned to its own answer. *Synthetic demonstration data.*

quietly.

7.2 The data-quality problems, and which were left open

Three source extracts arrive in the shape real ones do: a procurement extract, a technical/material extract and a PCF extract, each with its own naming and its own idea of a supplier identifier. Eight findings were logged (Table 5). Six were resolved deterministically through reference tables — never by fuzzy matching, so the resolution is reproducible and auditable. Two were left open *by design*, because they are the conditions the analytical layer must be able to survive: one material has no PCF at all, and one carries a PCF from an outdated reference year.

7.3 Pipeline and independent validation

The pipeline (Figure 4) is deterministic: given the same inputs it produces the same consolidated dataset and the same log. Its most important property is structural. A ground-truth version of the consolidated dataset exists in the repository, and the transform never reads it; validation is a separate step that compares the transform’s output against ground truth after the fact. A transform with access to the expected answer can be tuned until it agrees, which demonstrates nothing.

7.4 Semantic layer

The consolidated dataset is exposed through 21 audited measures. Three design decisions in that layer carry the method:

- **Baseline by case.** The incumbent baseline is resolved within the current case, so a delta is al-

Table 4. Thirteen practices across the decision sequence, with the strength of the evidence behind each. ■ strongly evidenced — primary standard or EU authority; □ supported — cited but narrower; △ project recommendation — plausible, not externally evidenced. The marker describes the support for the practice, never organisational maturity.

Step		Practice	Basis
SCREEN	□	Screen on hard constraints first, and keep what was screened out visible	Constraints vs objectives [18]
	□	Gate on specific legal or policy thresholds, not on general risk	[23, 5]
	□	Treat supply risk as a dimension, not a footnote	[8, 17]
COMPARE	■	Define equivalent function before any per-kg benchmark	[1] §A.1
	■	Satisfy the six comparability conditions before comparing suppliers	[1] §A.1, Table A.2
	□	Require the primary-data share, not merely a supplier figure	[3, 13]
	□	Declare a comparability verdict, not a confidence colour	[1, 3]
QUALIFY	■	Carry evidence quality beside the number, never inside it	[7, 3]
	■	Never substitute zero for missing environmental data	[7, 21]
DECIDE	□	Decide on total cost, not on price per kg	[6]
	□	Price in regulatory cost transmission before it arrives	[11, 10]
	□	Validate circularity against life-cycle impact	[2, 16]
RECORD	△	One decision record for procurement and engineering, not two	[6] and construction

ways against the right reference and never against a portfolio average.

- **A grain guard.** Percentage measures return blank unless a single material is in context. This makes a cross-case aggregate percentage structurally impossible rather than merely discouraged.
- **No imputation.** There is no coalesce-to-zero anywhere in the layer, so a missing PCF propagates as missing into every dependent measure and every visual.

One defect found during the audit is worth reporting because it is characteristic. Numeric columns had been typed without an explicit locale, and under a German-locale engine a value such as 240000.0 was parsed as 240,000,000 — a silent factor-of-ten class of error across several columns. It was corrected by making the locale explicit at every conversion step. The lesson generalises: in a benchmarking pipeline, a plausible wrong number is more dangerous than a missing one.

8 The analytics prototype

The semantic layer is displayed through a three-page interactive report: a portfolio overview, a material-level decision analysis and a portfolio screening matrix. Three behaviours distinguish it from a conventional supplier dashboard.

Gates are *shown, never applied as a filter*. A blocked alternative remains on every page with its block reason attached. A dashboard that filters out ineligible options hides the most valuable finding in the dataset, as Section 9 shows.

Evidence sits beside performance in its own columns — provenance, reference year and a data-quality tier — and is never folded into the performance figure. The tier is a prototype design heuristic derived from provenance and recency, and is labelled as such; it is not a standards-conformant data-quality assessment.

Missing values are rendered as missing. Where a carbon delta is undefined the cell reads n/a , in the same visual weight as any other cell, rather than as a zero or a gap in a bar.

9 Results

The deltas below demonstrate the method’s behaviour, not a validated supplier comparison. Measured against the six conditions of Section 5.2, the comparisons are at best conditional, as Section 12 sets out.

9.1 The portfolio total is not the sum of the options

The first result is arithmetic rather than environmental. Summing the annual spend of all 25 rows gives EUR 4,073,400. That figure is meaningless as a portfolio position, because the four alternatives in each case are mutually exclusive scenarios for the same demand: summing them counts that demand five times over. The position actually held is the incumbent portfolio — EUR 820,000 of annual spend, 480,000 kg of annual volume and 605,000 kg of annual CO₂e. The naive total overstates the portfolio by a factor of approximately five.

Table 5. The data-quality log. Six findings are resolved deterministically through reference tables; two are left open by design because they are the states the method must handle rather than remove. Leaving them open is what makes the missing-value and evidence-quality behaviour observable downstream. *Synthetic demonstration data.*

Finding	Source	Status	Resolution
Duplicate procurement record	Procurement	resolved	Vendor-code migration produced a second row for one material-supplier relationship; deduplicated against the technical source’s identifier
Material naming inconsistency	Procurement	resolved	Free-text description mapped to the canonical name by exact lookup
Material naming inconsistency	PCF	resolved	As above
Supplier naming inconsistency	PCF	resolved	Legal-form and spelling variants mapped to the canonical supplier
Country representation inconsistency	PCF	resolved	Country name and code variants mapped to a single representation
PCF unit basis inconsistency	PCF	resolved	Values reported per tonne converted to the per-kilogram basis before consolidation
Missing PCF value	PCF	open	One material has no reported PCF. Left missing: it is excluded from the carbon comparison and retained in every other
Outdated PCF reference year	PCF	open	One material carries a 2019 industry-average factor. Left in place and flagged, because suppressing it would hide the evidence problem

This is the kind of error that a scorecard with more columns does not prevent and a correctly specified baseline measure does. Coverage across the 25 rows is likewise reported rather than assumed: 23 of 25 alternatives are technically eligible (92.0%), 24 of 25 carry a PCF (96.0%), and 96.8% of spend is covered by a PCF.

9.2 All alternatives on one surface

Figure 5 places every alternative on one surface. The design carries three of the method’s commitments simultaneously. The two performance columns are visually dominant and diverge around the incumbent at zero. The four status columns recede to a pale tint when normal and become explicit only as exceptions — red for a technical blocker, amber for weak, outdated or missing evidence. And the missing carbon figure for SF-3003 is present as n/a , occupying its cell rather than vanishing from it.

Read as a screen, the surface answers one question per case rather than one question overall. That constraint is not a limitation of the visualisation; it is the analytical claim. It applies the same benchmarking principle as Figure 1 at portfolio scale: the dimensions stay separate, each is compared against its own reference, and no total is formed.

9.3 CASE-D: performance cannot buy eligibility

In the polymer additives case, the ranking on cost and carbon looks unambiguous. PA-4002 is 14.29% cheaper and 38.89% lower in CO₂e than the incumbent; PA-4004 is 7.14% cheaper and 16.67% lower. Both are technically blocked — PA-4002 is not approved, PA-4004 is under qualification — and neither

can be bought today.

The only eligible improvement is PA-4003, at −3.57% on cost and −5.56% on CO₂e: real, but a small fraction of the apparent opportunity. That, and not −14.29%, is the decision actually on the table.

Two consequences follow. First, performance cannot compensate for failed eligibility; a gate is passed or it is not. Second, the blocked options must stay visible with their reasons, because PA-4004 is under qualification and may return — which converts a screening result into a qualification-pipeline question with a date and an owner.

9.4 CASE-C: the best result on the weakest evidence

In the surfactants case all five materials are technically approved, so eligibility settles nothing and evidence settles everything (Table 6).

SF-3002 shows the strongest carbon result in this case at −54.76%. It rests on an industry-average factor from 2019, classified low confidence. The figure is not wrong; it is simply not yet a supplier comparison, because an industry-average factor cannot distinguish two suppliers inside the same industry.

SF-3003 is 8.33% cheaper and has no carbon figure at all. It remains blank on carbon in the measure, in the heatmap and in the case total. It is not zero, it is not a poor performer, and it is not dropped from the cost comparison. It is unknown, and it stays unknown.

		Cost Delta %	CO ₂ e Delta %	Technical eligibility	PCF available	PCF confidence	PCF reference year
CASE-A Base Oils	► BO-1001	0.00%	0.00%	✓	✓	✓	2023
	BO-1002	-12.50%	-20.00%	✓	✓	✓	2023
	BO-1003	-4.17%	+11.76%	✓	✓	✓	2023
	BO-1004	+8.33%	-35.29%	✓	✓	✓	2023
	BO-1005	+4.17%	+5.88%	✓	✓	✓	2023
CASE-B Solvents	► SV-2001	0.00%	0.00%	✓	✓	✓	2023
	SV-2002	-13.64%	+25.00%	✓	✓	✓	2023
	SV-2003	+22.73%	-35.71%	✓	✓	✓	2023
	SV-2004	-4.55%	+10.71%	✓	✓	✓	2023
	SV-2005	+7.27%	-14.29%	✓	✓	✓	2023
CASE-C Surfactants	► SF-3001	0.00%	0.00%	✓	✓	✓	2023
	SF-3002	+6.25%	-54.76%	✓	✓	low	2019
	SF-3003	-8.33%	n/a	✓	missing	no data	n/a
	SF-3004	+10.42%	-23.81%	✓	✓	✓	2023
	SF-3005	-2.08%	+7.14%	✓	✓	✓	2023
CASE-D Polymer Additives	► PA-4001	0.00%	0.00%	✓	✓	✓	2023
	PA-4002	-14.29%	-38.89%	X not approved	✓	✓	2023
	PA-4003	-3.57%	-5.56%	✓	✓	✓	2023
	PA-4004	-7.14%	-16.67%	X not approved	✓	✓	2023
	PA-4005	+3.57%	+8.33%	✓	✓	✓	2023
CASE-E Resins / Binders	► RB-5001	0.00%	0.00%	✓	✓	✓	2023
	RB-5002	-2.78%	+4.00%	✓	✓	✓	2023
	RB-5003	+2.78%	-5.60%	✓	✓	✓	2023
	RB-5004	-1.11%	-2.40%	✓	✓	✓	2023
	RB-5005	+1.11%	+2.40%	✓	✓	✓	2023

Figure 5. Portfolio screening across all 25 alternatives and five benchmark cases. Cost and CO₂e deltas are the analysis; technical eligibility and the three evidence columns are exceptions, shown recessed until they are not normal. Colour compares each material with the incumbent *of its own case*, and each case is an independent decision — the surface is a screen, not a league table. *n/a* means the value is unknown and is never read as zero; ► marks the incumbent. No score, no ranking and no weighting is derived. *Synthetic demonstration data.*

Table 6. CASE-C, all five materials. Performance and the confidence in that performance occupy separate columns, and the row with the strongest carbon result carries the weakest evidence. SF-3003 has no PCF: its carbon cells are empty rather than zero, and it remains in the cost comparison. *Synthetic demonstration data.*

Material	Cost Δ	CO ₂ e Δ	PCF	PCF data type	Year	Confidence
SF-3001	0.00%	0.00%	2.10	Supplier-specific (primary)	2023	High
SF-3002	+6.25%	−54.76%	0.95	Industry-average (secondary)	2019	Low
SF-3003	−8.33%	—	—	no PCF reported	—	No data
SF-3004	+10.42%	−23.81%	1.60	Supplier-specific (primary)	2023	High
SF-3005	−2.08%	+7.14%	2.25	Supplier-specific (primary)	2023	High

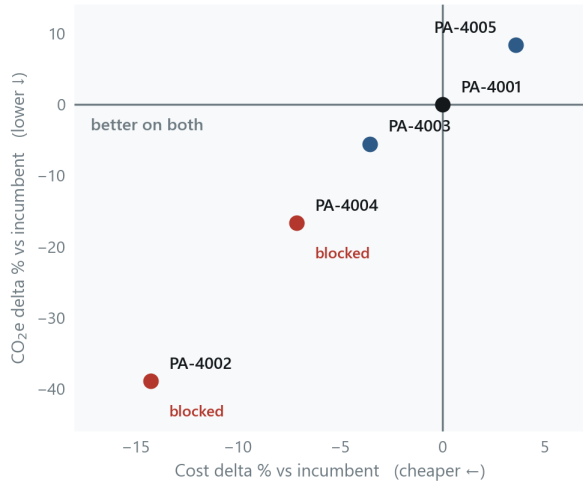


Figure 6. CASE-D on cost and carbon. The two alternatives furthest into the better-on-both quadrant, PA-4002 and PA-4004, are both technically blocked. The only eligible improvement is PA-4003, at a fraction of the apparent opportunity. *Synthetic demonstration data.*

10 Trade-offs and handling rules

10.1 Different tensions, different rules

The tensions in a sourcing decision are structurally different from one another (Table 7). Some resolve at a gate. Some are genuine two-axis trade-offs. One is about evidence rather than performance at all. Each has a handling rule, and not one of the rules is a weight.

This matters because weighting is a common reflex. A weight asserts an exchange rate between dimensions that the evidence does not supply, then hides that assertion inside a total. Where an exchange rate is genuinely needed — for instance a cost per tonne of CO₂e avoided — it should be displayed as a ratio so that it can be argued with, not embedded in a score.

10.2 Circularity is not automatically an improvement

Scope, stated before the finding: the following is a plastics waste-treatment study. It demonstrates a mechanism — burden shifting between impact categories — and is not evidence about base oils, solvents, surfactants, polymer additives or resins.

A European Commission JRC assessment [2] compared 27 recycling pathways across 14 impact categories. Recycling beat energy recovery on climate in every pathway analysed. On five other categories — acidification, particulate matter, ionising radiation, human toxicity (non-cancer) and eutrophication — energy recovery can perform better than energy-intensive recycling routes, and the assessment names those routes: PET alkaline hydrolysis, PS mechanical, MPO mechanical and pyrolysis, PE film mechanical and physical, and EPS physical recycling.

Three further findings from the same source matter for a benchmark method: technical quality issues in the recovered material; negative net incomes for methanolysis, pyrolysis and gasification; and, across the whole study, that “a clear ranking could not be established”. The result is also conditional on the European electricity mix, and the assessment states that as the mix decarbonises the gap widens further in favour of recycling. A circularity conclusion therefore carries a date.

The methodological consequence is direct. A universal ordering — impact first, circularity second, or the reverse — is not supported. Circularity and life-cycle impact are assessed as distinct outputs, hard constraints are applied first, and any priority rule is made explicit for the specific case rather than applied as policy.

11 Governance and decision ownership

Every practice in this paper fails at the same seam. Engineering owns the gates, procurement owns the commercial comparison, sustainability owns the evidence rules, compliance owns the legal thresholds — and each function keeps its own list. ISO 20400 addresses exactly this by placing sustainability criteria alongside price and quality inside one evaluation rather than in a parallel process [6].

The following ownership split and record design are a project synthesis and a design recommendation. ISO 20400 supports the principle; it does not validate this artefact.

Engineering defines the function and validates technical equivalence. Procurement owns commercial terms, capacity evidence and sourcing scenarios. Sustainability defines accepted methods and evidence requirements. Compliance defines the specific legal and

Table 7. The trade-off map. Eight tensions, each with the way it appears in a decision and the rule that handles it. Evidence strength uses the same three-state marker as Table 4. Cost–carbon quantitative evidence is steel-sector illustration only; no chemical-sector figure was retrieved, which is why that row carries the weakest marker.

Tension		How it appears	Handling rule
Cost ↔ carbon	△	The cheaper option carries more carbon, or the reverse	Two axes, plus a displayed cost-per-tonne ratio — never a score
Circularity ↔ non-climate impacts	■	Recovery wins on climate and loses elsewhere	Report as distinct outputs; any priority rule is stated per case
Circularity ↔ technical performance	□	Recycled content shifts the specification	Gate — recycled content is a candidate only if it passes equivalence
Circularity ↔ cost	□	The break-even arrives later than the target date	Treat the horizon as an explicit scenario, not an assumption
Sustainability ↔ technical performance	■	A carbon gain sits behind a failed approval	Gate. No carbon result buys past a technical blocker
Sustainability ↔ supply security	□	A carbon gain raises supplier concentration	State it beside the result; concentration needs awarded volumes
Performance ↔ evidence quality	■	The best number has the weakest provenance	Two columns. Never a quality-adjusted performance figure
Carbon ↔ other environmental burdens	■	A GHG-only view misses the shift	Disclose scope. Methods exist [24, 25]; the constraint is supplier data

policy gates and runs the due-diligence risk process. Suppliers provide declarations and evidence but never self-approve gate status. A named business decision owner accepts the residual trade-off, on the record.

The artefact that makes this operable is one decision record per benchmark case, carrying: incumbent and candidates; gate status with the reason, blocked options still visible; cost position and carbon position on separate axes; evidence status per number; the comparability verdict; the decision and what was given up; the named owner; and a review date. A second output falls out of the same record and should not be buried in it — the qualification pipeline: which blocked candidates are worth qualifying, by when, and at whose cost.

The prototype is not yet this artefact. Its material-level page is the closest existing approximation: it displays the decision context but captures neither the decision, nor the owner, nor the review date.

12 Scope boundary

The honest headline for the demonstrator is that its architecture is the right shape while the comparison it performs is not yet demonstrably valid — and that the conditions for making it valid are known and enumerable.

Measured against the six conditions of Section 5.2: the unit of analysis is a declared per-kilogram unit with no functional normalisation, although the dataset’s own active-content variation shows the materials are not functionally identical per kilogram; there is no boundary or data-period field; no methodology rulebook or version is recorded; provenance and a data-quality tier are present, but the primary-data share and any uncertainty statement are absent; representativeness is temporal only, through a reference year; and there is no verification field at all. **No condition is fully satisfied today.**

Eight fields would close the gap, and all eight are collected upstream in the specification and the request for quotation rather than repaired downstream in the analysis: a functional or declared unit with its reference flow; the methodology rulebook and version; a boundary description or comparability flag; the primary-data share; verification status and scope; the data or reference period; an uncertainty statement; and the comparability verdict itself.

Naming these is not an admission that weakens the demonstrator. It is what converts it from a picture into a specification.

13 Discussion

The recurring objection to a method without a score is that executives want one number. The evidence assembled here suggests that the single number is not available at the price usually assumed. A composite requires an exchange rate between dimensions; an EU authority running a full life-cycle and life-cycle-cost assessment across 27 pathways and 14 categories concluded that “a clear ranking could not be established” [2]. If that ranking could not be established with a complete inventory, it will not be established by weighting four columns of partially populated supplier data.

What a sequence buys instead is decision-grade failure. Each step produces a statement that can be argued with: this option is not eligible, and here is the threshold and its owner; these two figures are conditionally comparable, and here is the condition that is unmet; this improvement is real but rests on a 2019 industry-average factor. A composite converts all three into the same thing — a slightly lower score — and destroys the information that would have driven an action.

The demonstrator also shows that the expensive properties are structural rather than cosmetic. Gates

that screen without deleting, evidence carried beside performance, missing values preserved, and cases kept independent are all choices that are cheap to make at design time and very hard to retrofit once an aggregate exists. Adding a comparability verdict to this architecture is a data-collection task. Removing an embedded weighting from a deployed scorecard is an organisational one.

14 Limitations

Evidence limitations. No industrial chemical-procurement case supports the framework; the demonstration is synthetic throughout. No empirical study was found quantifying how far real supplier PCFs diverge when the §A.1 conditions are violated, so the size of the comparability problem is argued structurally rather than measured. TfS v3 and Catena-X v4 were not read in full, and are cited as examples of the additional programme specifications the standard requires, not as verified conformance targets. No chemical-sector green-premium dataset was retrieved; the cost-carbon exchange-rate evidence is steel-sector illustration only. No company-level supply-resilience metric set was retrieved. The GHG Protocol *Product Standard* dates from 2011 while a joint ISO/GHG Protocol product-standard process was reported active in 2026.

Access depth is disclosed per source in the reference list. Five primary sources were read in the original at section level; the majority of the remaining entries were read at summary level, and the paper’s claims are calibrated accordingly.

Method limitations. The four-role convention and the trade-off map are project syntheses. The three-state comparability verdict is a source-supported interpretation of the standard’s conditions, not a provision of the standard. The dimensions in Table 1 are a working set for this project’s material groups, not a fixed or universal list. Roles are not mutually exclusive: supply risk can be a gate, a performance dimension or context depending on the decision rule in force.

Demonstrator limitations. The prototype’s data-quality tier is a design heuristic derived from provenance and recency, not a standards-conformant data-quality assessment. Coverage percentages are computed against the project’s own inventory and weight unequal items equally. Supplier concentration is not derivable from the dataset, because mutually exclusive scenarios carry no awarded volumes. And, as Section 12 states, no §A.1 condition is fully satisfied.

15 Future work

Two build lanes follow, and one list is a refusal.

Lane A — make the existing comparison defensible. Add the eight fields of Section 12. This

stays inside the analytical question the prototype already asks and is the only work that changes the validity of its current output.

Lane B — add category modules where materiality justifies them. Recycled and renewable content with a chain-of-custody model [26]; due-diligence and human-rights performance; hazardous-substance legal status; criticality and supply resilience; water, land use, biodiversity, pollution and toxicity; and CBAM exposure where a product code falls outside current scope. Each is gated by the materiality question of Section 5.4 and each needs its own data owner and grain.

Do not build until real data exists. Qualification and switching cost; true total-cost effects on yield, energy, quality and downtime; supplier capacity, ramp-up and recovery time; country or supplier concentration on awarded shares; toxicity and legal use status for actual substances and applications; recycled or renewable chain-of-custody and mass-balance claims; and responsible-sourcing findings. Synthesising any of these would demonstrate columns rather than capability, and would corrupt exactly the credibility the rest of the method is built to protect.

16 Conclusion

This paper set out a method for comparing sustainable raw-material alternatives that does not produce a sustainability score. The method is a sequence: eligibility, then comparability, then evidence, then trade-off, then a recorded decision with a named owner. Ten success dimensions give the decision context a structured view, each read against a benchmark rather than combined. Four decision roles keep gates, performance, evidence and context from contaminating one another. Six standard conditions decide whether two supplier carbon figures may be compared at all, and a three-state verdict reports the answer.

Demonstrated on 25 synthetic alternatives across five independent cases, the method surfaced findings that a scorecard would have hidden: a portfolio total overstated approximately fivefold, two leading candidates that cannot be bought, a best-in-portfolio carbon result resting on a 2019 industry-average factor, and one material that has no carbon figure and is never given one.

The demonstrator does not prove comparability, and the paper says so and names the eight fields that would. That boundary is the contribution as much as the method is. A benchmark that reports what it cannot yet support is more useful to a sourcing decision than one that reports a number it cannot defend.

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Data and reproducibility. All data in this paper is synthetic. The dataset, the deterministic transformation and validation scripts, the semantic-layer measure definitions and the generators for every figure in this paper will be published with the project repository. Every source used is listed above with the access depth at which it was read. No supplier, company or market data is used or implied anywhere in the project.
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